TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Determination of Spirality and Twisting Percentage

The skew that twists a T-shirt after washing: measuring wale displacement in knitted fabric, converting it to spirality percentage and angle, and tracing it back to the yarn.

TE-3202 · QUALITY EVALUATION OF FABRIC AND APPAREL SESSIONAL · EXPERIMENT 13 · LEVEL 3 TERM 2

1. Aim and theory

Aim: to determine the spirality (torque) of a knitted fabric after washing, by the displacement method used in the AATCC 179 / ISO 16322 family of methods for spirality and torque in knitted fabrics and garments, expressing the result as spirality percentage and spirality angle, identifying the causes, and judging the fabric against a quality limit before and after remedial finishing.

1.1 What spirality is

Spirality is the twisting distortion of a knitted structure in which the wales — which should run squarely down the fabric, perpendicular to the courses — lean sideways after relaxation, so that side seams of a garment rotate toward the front or back, hems skew, and stripes describe a shallow helix around the body. It is the knitted fabric's characteristic dimensional fault, distinct from simple shrinkage: the fabric may keep its area almost unchanged while its geometry shears. A T-shirt whose side seam arrives at the front after three home washes has not shrunk crookedly by accident; it has expressed a torque that was built into it at knitting and released by washing.

1.2 Where the torque comes from

Three causes combine, and the viva expects all three. First, **twist-lively yarn**: a singles spun yarn carries residual torque from its twist — it wants to untwist — and in the relative freedom of a knitted loop that torque tilts every loop in the same rotational sense, skewing the whole fabric one way. Z-twist and S-twist yarns spiral in opposite directions, which is why plying, or alternating twist direction between feeders, is a structural cure. Second, **single-jersey asymmetry**: the plain single-jersey loop is inherently unbalanced — the loop legs and sinker loops pull unevenly — so single jersey shows the strongest spirality of the common structures, while balanced interlock and rib constructions largely cancel their own torques course by course. Third, **finishing and making-up**: a skewed fabric can be pulled square on the stenter and held there by resin or heat setting only as a loan — the first thorough

wash calls the loan in — while cutting panels from an already spiralled fabric bakes the twist into the garment permanently. Heat setting is a genuine stabiliser for thermoplastic fibres (polyester jersey can be set square), and resin or compacting treatments reduce movement in cotton, but none of them removes a large yarn torque; they restrain it, and the test after washing reports how well.

1.3 Measurement

The fabric method of the AATCC 179 / ISO 16322 family marks a square on the conditioned fabric with one side along a wale, washes and dries the specimen by the specified procedure, then measures how far the wale line has displaced sideways over a known course-wise length. Let the original wale line, extended over a measured length Y along the fabric, finish a horizontal distance X away from its true square position after washing. Then:

$$\text{Spirality (\%)} = X / Y \times 100$$

$$\text{Spirality angle } \theta = \tan^{-1}(X / Y) \quad \cdot \quad \text{for small angles, } \theta \text{ (degrees)} \approx \text{spirality (\%)} \times 0.573$$

The percentage is the commercial language — buyer specifications for knitted garments commonly set a limit around 5% spirality after washing (the exact figure is always the buyer's own) — while the angle is the garment-maker's language, since pattern displacement and seam rotation are angular events. Whole-garment methods in the same family wash the made garment and measure seam displacement directly; the fabric method here predicts that behaviour earlier and more cheaply, at the fabric stage where correction is still possible. As with dimensional stability, specimens are conditioned and measured in the standard atmosphere (65% RH, 20 °C), laid flat without tension, and the washing procedure is named in full in the report.

1.4 Remedies, in order of honesty

Prevention outranks cure. Balanced yarns (plied, or low in residual torque by steam setting of the yarn) and balanced structures remove the cause. Twist management at spinning — lower twist multiplier where strength allows, twist setting — reduces the driving force. At finishing, compacting gives the fabric its relaxed geometry mechanically, heat setting locks thermoplastic structures in their square state, and resin finishing restrains cotton jersey at some cost to handle and strength. At cutting, panels can be cut to the spiralled geometry as a last resort so the garment, not the fabric, is square — a legitimate salvage that must be declared, because it wastes cloth and fails if the consumer's washing moves the fabric further. The re-test in the worked example shows the finishing route succeeding on this fabric; the laboratory's duty is to verify the improvement after washing, not to accept the finisher's intention.

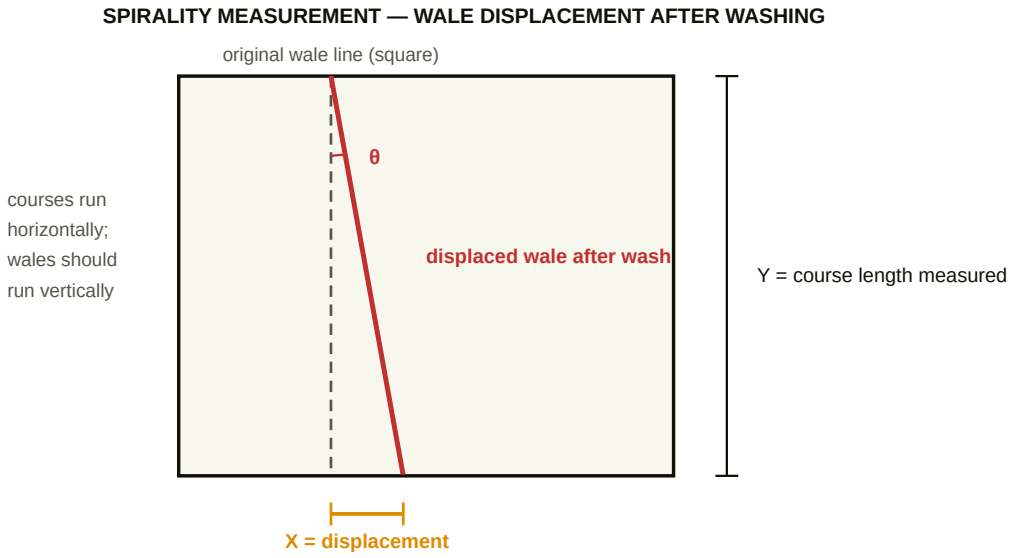


Figure 1 — Measuring spirality on a washed square specimen. The wale that left the mark vertically (dashed) is displaced sideways by X over the measured length Y (here drawn exaggerated for clarity). Spirality % = X/Y × 100 and θ = tan−1(X/Y).

2. Apparatus and procedure

- **Apparatus:** washing equipment and procedure as specified (domestic washing procedure, named in the report); indelible marker and square template; steel rule (mm) and a square or protractor for the angle check; flat measuring table; conditioning facilities (65% RH, 20 °C).
- **Marking:** condition the fabric, lay it flat and relaxed, and mark a square of known size with one edge exactly along a wale and one along a course; draw the reference wale line clearly along its full length. Record the length Y over which displacement will be measured (500 mm in this exercise, using a suitably large specimen).
- **Washing:** subject the specimen to the specified washing and drying cycles (three cycles in this exercise, matching the dimensional-stability work of Experiment 12), then condition it again in the standard atmosphere (65% RH, 20 °C).
- **Measurement:** lay the specimen flat without tension, extend the displaced wale line, and measure X — the sideways displacement of the wale from its square reference position at the end of length Y — to the nearest millimetre. Measure at more than one wale if the specimen allows and average. Optionally confirm the angle directly with a protractor against the calculated angle.
- **Calculation and judgement:** compute spirality percentage and angle; state the direction of spiral (clockwise or anticlockwise as viewed) since remedy and cutting depend on it; judge against the stated quality limit; where a remedial finish is applied, repeat the whole sequence — mark, wash, measure — on a fresh specimen of the finished fabric and report both results.

3. Worked calculation

Cotton single-jersey specimen, washed by the stated procedure. Measured length Y = 500 mm; sideways displacement of the marked wale after washing X = 38 mm.

$$\text{Spirality} = X / Y \times 100 = 38 / 500 \times 100 = 7.6\%$$

$$\theta = \tan^{-1}(38 / 500) = \tan^{-1}(0.076) = 4.35^\circ$$

Against a quality limit of 5% spirality, the fabric **fails**: at 7.6% a garment's side seam would rotate by roughly 38 mm over every half-metre of length — plainly visible on a T-shirt, and enough to take striped patterns visibly off-square. The magnitude and the single-jersey cotton construction point to the classic cause, a twist-lively singles yarn expressing its torque on first washing.

Re-test after remedial finishing. A fresh specimen of the same fabric, resin-finished and heat-set under the finisher's conditions, was marked, washed identically and re-measured: X = 22 mm over the same Y = 500 mm.

$$\text{Spirality after finishing} = 22 / 500 \times 100 = 4.4\% \quad (\theta = \tan^{-1}(0.044) \approx 2.52^\circ) \text{ — PASS against the 5\% limit}$$

RESULT

Before finishing: X = 38 mm, Y = 500 mm → spirality 7.6%, θ = 4.35° — fails the 5% limit. After resin/heat-set finishing: X = 22 mm → 4.4% — passes. The finish restrains about two-fifths of the displacement ((38 − 22) / 38 × 100 ≈ 42%); the yarn torque itself remains, so the pass is conditional on the finish surviving the consumer's washing — durability re-tests govern bulk acceptance.

4. Precautions

- Mark the reference wale on the relaxed, conditioned fabric before washing, following one true wale. A reference line drawn along a yarn path that already wanders builds the wander into every later measurement.
- Measure X at the full stated length Y, at right angles to the reference direction. A short Y magnifies reading error into percentage error — 1 mm misread is 0.2% at 500 mm but 1% at 100 mm.
- Do not pull the washed specimen square on the table "to measure it properly". The displacement is the result; flattening for measurement means laying flat without tension, not correcting the skew by hand.
- Wash exactly as specified and report the procedure and cycle count. Spirality grows over the first washes as torque releases; a single gentle cycle understates the garment's eventual twist.

- Record the direction of the spiral, not only its size. Two fabrics at 5% in opposite directions need opposite yarn or cutting remedies, and a report without direction sends the mill to fix the wrong one.
- Verify remedial claims by full re-test on fresh specimens of the finished fabric, washed identically. A finish that measures well unwashed has demonstrated nothing — spirality is defined after washing.

Practice (viva preparation)

1. Define spirality and distinguish it from shrinkage.
2. Give the three causes of spirality in knitted fabric.
3. A wale displaces X = 38 mm over Y = 500 mm. Find the spirality percentage and the angle.
4. Judge the result of Question 3 against a 5% quality limit and describe the garment fault it predicts.
5. After finishing, the same fabric gives X = 22 mm over 500 mm. Find the new spirality and the verdict.
6. Why is single jersey more prone to spirality than interlock?
7. Why must a remedial finish be judged by a washed re-test rather than by measuring the finished fabric dry?

Answer key

1. Spirality is the sideways skewing of the wales of a knitted fabric after relaxation and washing, caused by torque in the structure, so courses and wales are no longer perpendicular. Shrinkage is a change of dimensions (length and width); spirality is a change of geometry — a fabric can spiral strongly with almost no net shrinkage, the area shearing rather than contracting.
2. (1) Twist-lively singles yarn whose residual torque tilts every loop the same way. (2) The inherent asymmetry of the single-jersey loop, which does not balance its own pulls. (3) Finishing and making-up that hold the fabric artificially square (stenter, resin) until washing releases it, or that cut garments from already-skewed fabric.
3. Spirality = 38 / 500 × 100 = 7.6%. Angle $\theta = \tan^{-1}(38/500) = \tan^{-1}(0.076) = 4.35°$.
4. 7.6% exceeds the 5% limit — fail. In a garment, a side seam would rotate about 38 mm toward the front or back over a 500 mm length, hems and stripes would sit visibly skewed, and the fault worsens with the first washes the customer gives it.
5. Spirality = 22 / 500 × 100 = 4.4%, inside the 5% limit — pass. The finish has restrained the displacement by 16 mm of the original 38 mm (about 42%), though the underlying yarn torque persists beneath the restraint.
6. Single jersey is knitted from one yarn system forming an unbalanced loop: the legs and sinker portions pull unequally and every course adds torque in the same sense, with no opposing structure.

Interlock is built from two intermeshed rib systems whose loop torques oppose and cancel course by course, so the relaxed structure stays square even with lively yarns.

7. Because the fault itself only appears on washing: torque held by a dry finish or a stenter is invisible until water, heat and agitation release it. A dry measurement certifies the finisher's setting, not the garment's behaviour. Only a specimen marked, washed by the specified procedure and then measured shows the spirality the wearer will see — and repeated cycles show whether the restraint itself survives laundering.
